F78AA1 – ACTUARIAL AND FINANCIAL MATHEMATICS A
December 2009

Exam

The exam was straightforward and not as time pressured as in previous years but because the style of some of the questions was different from recent years, some students felt the exam was difficult. However, students with a good understanding of the subject did not have any problems and a number of students achieved very high marks.

Q1 and Q2 were standard bookwork. All students should have been able to score nearly full marks if they had learned the notation and standard methods required for this course. It is essential that this notation is well learned, as many actuarial calculations are based on this notation, and students need to become totally at home with it as soon as possible. If not, they will struggle with the majority of their actuarial exams.

Some students were confused in part 2(i)b because the rate of increase of the annuity was higher than the interest rate. However, the algebra is the same as for the usual case, which all students should be very familiar with. Even if not, the solution can be calculated from first principles using a geometric progression.

Part 2(i)c was quite a complicated calculation, but with knowledge of the increasing annuity function, $(Ia)_x$ should not have presented any problem.

Q3 was a standard question on bonds, albeit worded differently from some of the past exam questions.

Q4 confused a lot of students because it referred to an interest-only loan. But, in fact, the calculations in 4(a) and 4(b) are much simpler than the standard question (ie where equal repayments of capital and interest are made). However, students needed to understand how interest is applied to work out how to answer this question. This question was poorly answered by the majority of students.

Q5 was again a standard question and one that has been asked in previous years (with different numbers) apart from part(c). Students who had looked at past papers should all have scored 6 or more (out of 8) on this question.

A few questions were interpreted differently from expected, but full marks were given for either interpretation. These were 1(i) where the time period was ambiguous, 2(iii) either £1,000 was invested at the start of year 1 only, or £1,000 was invested at the start of year 1 and another £1,000 at the start of year 2, and 3(c) where the redemption proceeds were taken as 120% nominal x inflation growth rather than the usual inflation growth only.

Exam style:

- If you are ever in doubt about how to interpret a question, clarify what you understand is being asked before you start on the calculation. This lets the examiner know why you might not have answered in a standard way, and you could even get credit for an incorrect answer.
- Do not fudge your answers; the examiner will easily spot this and you could lose marks

- Try to write tidily. Use double spacing if you think you might go back to amend what you have written.

Project

The project was generally well done.

Some students probably spent more time on it than they needed given the number of marks available.

The main criticism was that not all students answered in the style required, namely that this was a report written to an investor giving him a recommendation. Hence, it is important to explain everything in normal English, and to give a clear recommendation with reasons. The investor has come to you for a decision on what to do, not a range of options.

In terms of the calculation, some students incorrectly used the NPV and IRR functions in Excel for the half-yearly payments and this gave incorrect answers.

It was also disappointing how few students related the yield to the UK economy at the moment. Several students said, for example, that if interest rates were above 15% they should invest in A otherwise B. But we know that interest rates are currently low and are expected to remain low in the short term. Consequently a better answer might have been to say something like "invest in B unless you believe that interest rates will rise to above 15% over the next 10 to 20 years".

As ever, it is key to get in as many points as possible. Some students focused too much on one idea and so scored less than they could have done.

General Comments

This course, Financial and Actuarial Mathematics, and Life Insurance Mathematics underpin all actuarial work.

For students to succeed they must invest some time in understanding these courses totally. A couple of weeks' hard revision might get you a pass, but will leave you struggling through the rest of your exams. Consistent work throughout the semester will embed the fundamental principles and give you much more chance of getting exemptions, and will enable you to understand the lectures better.

Spending time on the tutorial questions before the tutorial itself, and reviewing the lecture material each week will really make a big difference to your understanding of the subject, and will in the end save you time.

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